Performance Evaluation of Traffic Steering Methods for QoS Improvement in case of Network Failures in Beyond 5G Networks

Koki Kawaguchi, Daiki Nobayashi, Kazuya Tsukamoto, Takeshi Ikenaga

Abstract Mobile communication networks are expanding. However, when natural disasters happen and communication equipment is damaged, service disruptions can occur and many people cannot use the network. When such disasters happen, it is possible to use long-distance wireless networks such as Low Power Wide Area (LPWA) networks or satellite communication networks as alternative routes. This study focuses on the network between the Beyond 5G core network and the remote host and proposes a resilient network that uses the open RAN (O-RAN) architecture to utilize available communication networks during failures. Additionally, we propose a traffic steering method based on the types of user devices to effectively use available alternative routes and alleviate QoS degradation for users. This paper evaluates the effectiveness of the proposed system through simulations. The simulation results showed that the proposed methods increase the satisfaction rate of flows' requirements.

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1 Introduction

As 5G networks expand, mobile networks are becoming indispensable. However, there are a lot of natural disasters such as earthquakes, typhoons, and heavy rains in Japan. These natural disasters can damage communication equipment and cause cable breaks, leading to communication disruptions. For example, the 2011 Great East Japan Earthquake caused severe damage to NTT East's communication infrastructure, resulting in widespread service outages across many regions[1]. During such disasters, it is possible to use long-distance wireless networks such as LPWA networks or satellite communication networks as alternative routes. For example, K. Ali et al. [2] examined the utilization of emerging communication technologies such as IoT, device-to-device communications, and cloud computing to maintain communication during disasters. There is a demand for networks that remain connected or at least can quickly recover connections during disasters. Therefore, this study aims to maintain network connectivity and alleviate the degradation of Quality of Service (QoS) in the event of communication failures caused by natural disasters.

This study proposes a resilient network for Beyond 5G network that uses the open RAN (O-RAN) architecture [3] to leverage available communication networks during failures. Additionally, we propose a traffic steering method to effectively use finite communication resources and alleviate the QoS degradation for users. This study evaluates the effectiveness of the proposed resilient network system and the flow allocation method for User Equipment (UE). We compare the flow allocation method to a round-robin allocation of flows from each UE to available wireless communication networks. Using the network simulator ns-3, this study demonstrates that the proposed method increases the satisfaction rate of user applications that meet their required QoS.

The significance of this research lies in its potential to enhance the resilience and reliability of future mobile networks. By addressing the critical issue of maintaining communication during natural disasters, this study contributes to the ongoing efforts to develop more robust and adaptive network infrastructures. Moreover, the integration of multiple communication technologies and the use of advanced traffic steering algorithms represent a forward-thinking approach to network management in the Beyond 5G era.

The rest of this paper is organized as follows. Section 2 reviews related studies on traffic steering in 5G networks and resilient network for natural disasters. Section 3 explains the details of the proposed method. Section 4 describes the simulation environment constructed for the evaluation. Section 5 provides the evaluation results and discussion. Finally, Section 6 concludes the paper.

2 Related Works

This section introduces related research on traffic steering within the O-RAN architecture and resilient networks for natural disasters.

F. Kavehmadavani et al. [4] proposed a framework called JIFDR, which integrates traffic prediction, flow-split, dynamic user association, and radio resource management to address unknown dynamic traffic demands within the O-RAN architecture. Both F. Kavehmadavani et al. and this study leverage the O-RAN architecture to manage dynamic traffic demands. However, there are differences as follows. F. Kavehmadavani et al. used a long-short-term memory (LSTM) model for traffic prediction and steering. This study focuses on traffic steering based on the applications used in UE.

Next, we examine related studies on traffic steering between 3GPP access networks such as 4G and 5G, and non-3GPP access networks such as Wi-Fi. T. Cogalan et al. [5] proposed a method that measures the real-time link status and offloads data to non-3GPP networks to reduce the network load. Both T. Cogalan et al. and this study use multiple wireless networks for traffic steering. However, there are differences as follows. T. Cogalan et al. used simultaneously multiple access networks to send data. This study allocates flows to a single route.

Additionally, K. Ali et al. [2] discussed the importance of IoT and 5G networks during natural disasters. The paper examined how to utilize emerging communication technologies such as IoT, device-to-device communications, vehicular networks, cloud and fog computing, unmanned aerial vehicles, and sensor networks for communication during disasters. Both K. Ali et al. and this study propose resilient communication networks for natural disasters. However, there are differences as follows. K. Ali et al. focused on a broader range of technologies and mainly addresses the access network. In contrast, this study specifically targets the network between the Beyond 5G core network and the remote host, proposing a resilient network system using the O-RAN architecture and a traffic steering method based on user device applications.

3 Proposed Method

This study proposes a resilient network system that utilizes O-RAN and network slicing within Beyond 5G networks. The system is designed to flexibly use available communication routes during failures and effectively manage finite communication resources to alleviate QoS degradation for users. In this section, we introduce the details of the proposed system and the traffic steering method.

3.1 System of Resilient Network

Fig. 1 illustrates the proposed network system. In this system, in addition to the Beyond 5G network, available wireless communication networks such as Low Earth Orbit (LEO) satellite services, Geostationary (GEO) satellite services, and Low Power Wide Area Networks (LPWAN) are utilized as slices in the Beyond 5G

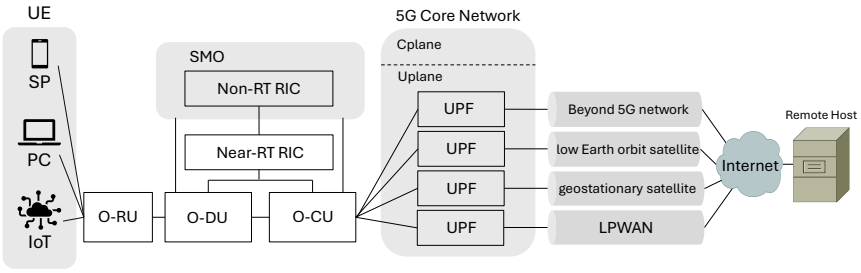


Fig. 1: Proposed Architecture.

framework. This system assumes the use of LoRa for LPWA. The proposed system allows the construction of a network that can flexibly use multiple communication paths. Under normal conditions, the system uses the Beyond 5G network, which provides the largest bandwidth. The following sections describe the operation of the system under normal conditions.

1. The non-real-time RAN intelligent controller (Non-RT RIC) stores the required throughput and latency for the applications used by each UE.
2. The Non-RT RIC also stores the bandwidth and latency data of wireless communication networks such as LEO satellite services, GEO satellite services, and LPWAN.

When a natural disaster damages communication equipment and the Beyond 5G network becomes unavailable, the proposed system maintains UE communication by using other available routes. However, these alternative routes may not provide enough bandwidth and latency as well as the Beyond 5G network. Therefore, it is necessary to allocate UE flows appropriately to the finite communication resources. The proposed system mitigates QoS degradation by allocating each flow based on the throughput and latency requirements of the applications used by the UEs. The system operates in the following steps and allocates flows for each UE to the alternative route:

1. The Non-RT RIC detects a communication failure.
2. When a UE connects to the 5G network, the Non-RT RIC sends the required throughput and latency for the applications used by the UE to the near-real-time RIC (Near-RT RIC). Additionally, the Non-RT RIC provides the bandwidth and latency of available wireless communication resources to the Near-RT RIC.
3. The Near-RT RIC determines the routing path for the flow and notifies the O-RAN Central Unit (O-CU) of the path.
4. The O-CU identifies the packets based on the UE's IP address and directs them to the specified route.

Table 1: Link Configuration

Route	Bandwidth	MTU	Latency
LEO satellite	130 Mb/s	1500 B	50 ms
GEO satellite	120 Mb/s	1500 B	800 ms
LPWA	50 kb/s	255 B	5 s

3.2 Traffic Steering Method

This section describes the flow allocation algorithm used by the system explained in 3.1. The three alternative routes to the Beyond 5G network considered are LEO satellite services, GEO satellite services, and LPWAN. The allocation of UE flows is based on the required throughput and latency of the applications used by the UEs. The algorithm shown in Fig. 2 determines the path for each flow. Note that A and B in Fig. 2 represent either LEO satellite services or GEO satellite services.

When a new UE connects to the 5G network, the system calculates the total required throughput by summing the required throughput of the flows currently allocated to LPWAN and the new UE's required throughput. If the total does not exceed LPWAN's bandwidth and the required latency is longer than LPWAN's latency, the flow is assigned to LPWAN. For the remaining flows, they are first categorized by latency and then by throughput. If the sum of the required throughput values of the flows assigned to a path exceeds the path's bandwidth, the system checks the condition of the other path and makes the final allocation accordingly.

4 Performance Evaluation

This section describes the simulation parameters, the metrics used for assessment, and the methods used for comparison. We create a topology the same as shown in Figure 1 in the network simulator ns-3. Alternative routes are LEO satellite services, GEO satellite services, and LPWAN, following the proposed method. The performance settings for each route are summarized in Table 1. Parameters in Table 1 are based on [6]-[8].

Additionally, each UE uses one of the ten application scenarios. Table 2 shows the transmission rates, required throughput, and required latency for each application. All applications used UDP as the transport protocol. The number of UEs using each application was randomly set between seven and ten. The blank fields indicate applications with no specific latency requirements in Table 2. However, the proposed system allocates flows of Video Streaming I and Video Streaming II to the available alternative routes with the minimum latency.

We randomly set each UE's transmission start time within the range of 0.0 to 1.0 seconds from the beginning of the simulation. Therefore, the order of UE con-

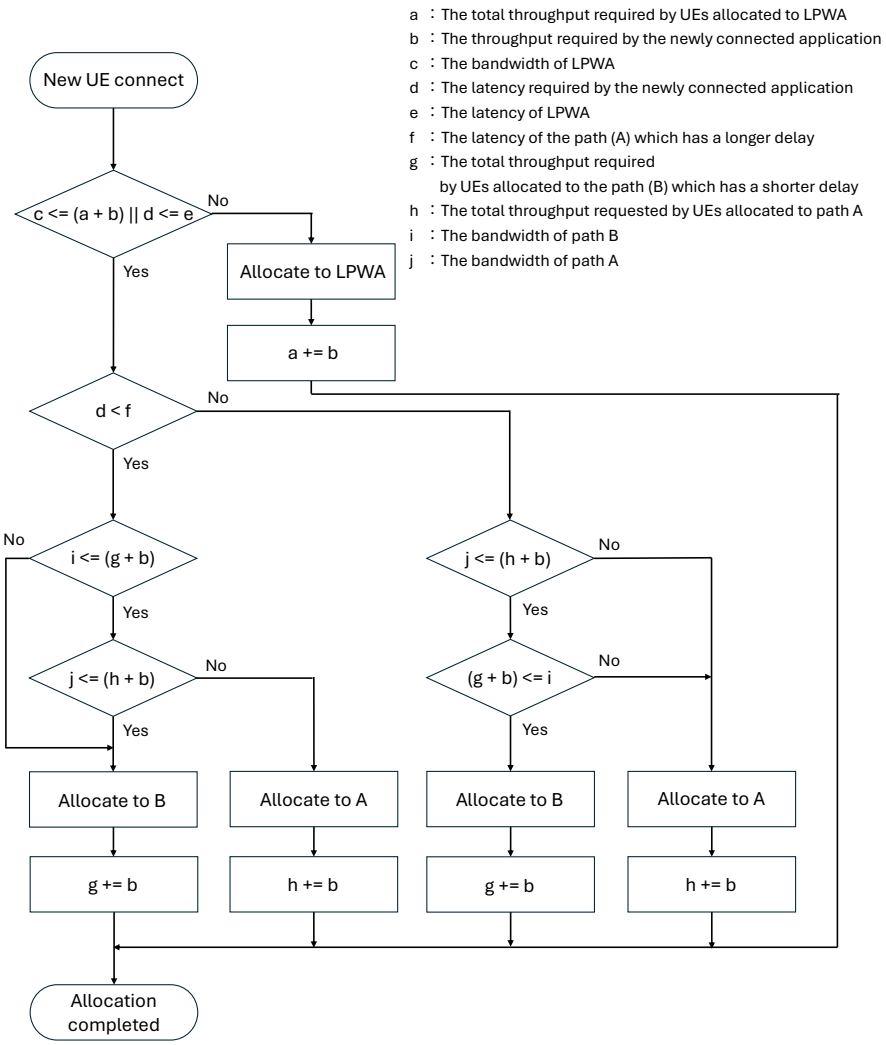


Fig. 2: Flow Chart of Flow Allocation.

nections to the 5G network was random. Additionally, each UE transmits the data for 14 seconds. We evaluate the simulation results based on the satisfaction rate of flows that met the required throughput and latency. For applications without specific latency requirements, we use the average latency of each application for evaluation.

Table 2: Application Configuration

Application	Required throughput	Required latency	Sending rate
Video Streaming Ⅰ [9]	6.0 Mb/s	-	6.6 Mb/s
Video Streaming Ⅱ [9]	4.0 Mb/s	-	4.4 Mb/s
Call [10]-[11]	64 kb/s	150 ms	70 kb/s
Gaming [12]-[13]	6.0 Mb/s	100 ms	6.6 Mb/s
File upload [14]	2.52 Mb/s	-	2.8 Mb/s
IoT Ⅰ [15]-[16]	0.8 kb/s	-	0.9 kb/s
IoT Ⅱ [15]-[16]	0.7 kb/s	-	0.8 kb/s
Video call Ⅰ [17]-[19]	3.8 Mb/s	150 ms	4.2 Mb/s
Video call Ⅱ [17]-[19]	2.5 Mb/s	200 ms	2.7 Mb/s
V2N [20]	2.5 Mb/s	1000 ms	2.7 Mb/s

To evaluate the proposed flow allocation method, we used the following allocation method as a comparison method.

1. All flows of IoT devices were assigned to LPWA. Since LPWA meets the requirements of IoT devices[8].
2. For non-IoT device flows, we used a round-robin algorithm, alternating flows between LEO satellite services and GEO satellite services in the order they were generated.

5 Simulation Results

We simulated three times for both the proposed method and the comparison method. Figure 3 shows the simulation results for the satisfaction rate of flows that met the required throughput and latency for each application used by the UEs. Additionally, Figure 4a shows the average latency for flows requiring shorter latency, and Figure 4b shows the average latency for flows without specific latency requirements.

Compared to the comparison method, the proposed method increased the satisfaction rate of flows meeting the required throughput by 15.1 points and the required latency by 30.3 points. The uneven distribution of data across the flows led to an 87.4 % satisfaction rate for the required throughput, rather than a uniform reduction in throughput across all flows. The 85.2 % satisfaction rate for the required latency resulted from some flows being assigned to GEO satellite paths. This occurred because the LEO satellite paths reached their capacity, despite those flows requiring shorter latency.

Additionally, the proposed method achieved a shorter average latency for flows requiring low latency. For example, in the case of Video Streaming I, the average latency decreased from 420.3 ms to 382.9 ms with the proposed method. In contrast, the latency increased for File uploads. The comparison method allocated the path to either LEO satellite communication or GEO satellite route, while the pro-

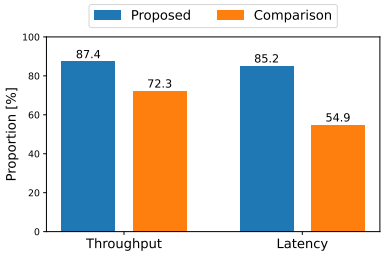
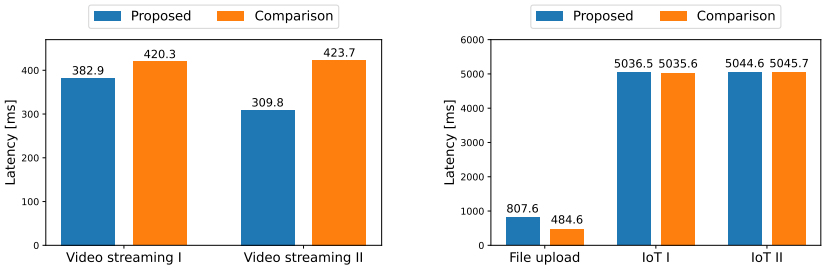


Fig. 3: Satisfaction Rate.



(a) Flows Requiring Shorter Latency.

(b) Flows Requiring No Specific Latency.

Fig. 4: Latency.

posed method primarily used GEO satellite paths. Regarding IoT applications, both methods allocated the path to LPWA, leading to minimal changes in latency.

Therefore, we consider it possible to maintain communication and alleviate the degradation of QoS by using alternative routes and appropriately allocating flows during network failures.

6 Conclusion

In this paper, we proposed a resilient network system that flexibly leverages available communication networks during network failures. Additionally, we propose a traffic steering method to efficiently use finite communication resources and alleviate QoS degradation for users. We simulated to verify the effectiveness of the proposed method. Simulation results showed that the proposed method increased the

satisfaction rate of flows meeting throughput and latency requirements compared to the round-robin algorithm by steering traffic based on application needs.

A key future challenge is addressing scalability issues caused by different network environments and an increasing number of users. We plan to conduct further performance evaluations in real operational environments and tests with more diverse scenarios.

Acknowledgements This paper was partly supported by JSPS KAKENHI Grant Number 23K11067.

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